

CSS Fundamentals

Syntax & Selectors · Colors & Backgrounds · Typography · Borders & Spacing · Box Model

CSS Syntax

Colors & Backgrounds

Typography

Borders & Spacing

Box Model

Prepared & Presented by

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Technical Trainer | Frontend Development Bootcamp 2026

About the Presenter

Utkarsh Kushwaha

Full Stack Developer | Technical Speaker | Community Mentor

Experienced in building responsive and interactive web applications using modern technologies and actively involved in technical communities, workshops, and student learning initiatives.

TECHNICAL SKILLS



HTML5



CSS3



JavaScript



React.js



Node.js

express

Express.js



MongoDB



Git & GitHub



REST APIs



Responsive
Web Design

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Learning Objectives

By the End of This Session, Students Will Be Able To:



Understand the purpose and importance of CSS



Write correct CSS syntax using selectors



Apply colors and backgrounds to elements



Style text using typography properties



Control spacing using borders, margin & padding



Understand and apply the CSS Box Model

Session Outcome: Students will gain a practical understanding of CSS fundamentals used to style real-world webpages.

Session Agenda



Introduction to CSS

What CSS is & why it matters



CSS Syntax & Selectors

Rules, properties, selectors



Colors & Backgrounds

Color formats, bg properties



Typography & Text Styling

Fonts, size, spacing, align



Borders & Spacing

Border, margin, padding



CSS Box Model

Content, padding, border, margin

Learning Outcome: Apply styling and visual enhancements to web pages using CSS.

Quick Revision

Previously Learned in Session 2

- HTML Tables (rows, columns, merge cells)
- HTML Forms & Input Elements
- Labels & Form Controls
- Semantic HTML Tags
- Audio & Video Embedding
- iframe Embedding

CODE

```
<table border="1">
  <tr><th>Name</th><th>Age</th></tr>
  <tr><td>Rahul</td><td>20</td></tr>
</table>
```

```
<form>
  <label>Email:</label>
  <input type="email">
</form>
```

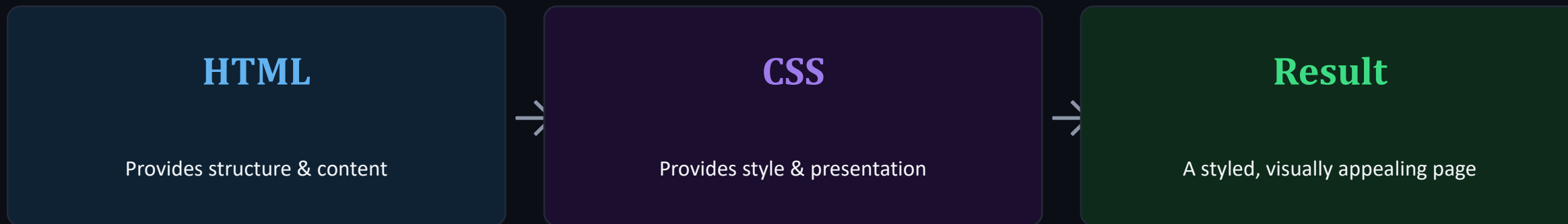
```
<video controls>
  <source src="movie.mp4">
</video>
```

Introduction to CSS

Why We Style Webpages

What is CSS?

CSS (Cascading Style Sheets) controls how HTML elements look on a webpage.



CODE

```
<!-- HTML -->
<h1>Welcome to CSS</h1>

/* CSS */
h1 {
  color: blue;
  font-size: 32px;
}
```

- ✓ Separates content from design
- ✓ Reusable across multiple pages
- ✓ Makes sites responsive & attractive
- ✓ Reduces repeated styling code

Ways to Add CSS

Three Methods to Apply Styles to HTML

Inline CSS

Written directly inside an HTML tag using the style attribute.

```
<p style="color:red;">
  Hello
</p>
```

Internal CSS

Written inside a <style> tag in the <head> section.

```
<style>
  p { color: red; }
</style>
```

External CSS

Written in a separate .css file and linked using <link>.

```
<link rel="stylesheet"
      href="style.css">
```


CSS Syntax & Selectors

Rules, Properties & Targeting Elements

CSS Syntax

A CSS rule is made up of a selector and a declaration block.

CODE

```
selector {  
  property: value;  
  property: value;  
}  
  
p {  
  color: navy;  
  font-size: 18px;  
  text-align: center;  
}
```

selector

Targets the HTML element(s) to style

property

The style aspect you want to change

value

Setting applied to that property

{ }

Declaration block wraps all rules

Types of CSS Selectors

Element Selector

```
p { color: red; }
```

Targets all elements of a given tag

Class Selector

```
.highlight { ... }
```

Targets elements with a class attribute

ID Selector

```
#header { ... }
```

Targets a single unique element by id

Universal Selector

```
* { margin: 0; }
```

Targets every element on the page

Group Selector

```
h1, h2, p { ... }
```

Applies the same rule to multiple selectors

Descendant Selector

```
div p { ... }
```

Targets elements nested inside another



Colors & Backgrounds

Bringing Visual Identity to Webpages

CSS Color Formats

Colors can be defined in several formats in CSS.



Named Colors

```
color: red;
```



HEX Code

```
color: #FF5733;
```



RGB

```
color: rgb(255,87,51);
```



RGBA

```
color: rgba(255,87,51,0.6);
```



HSL

```
color: hsl(9,100%,60%);
```

Background Properties

CODE

```
div {  
  background-color: #2496ED;  
  background-image:  
    url('bg.jpg');  
  background-repeat: no-repeat;  
  background-position: center;  
  background-size: cover;  
}
```

background-color

Sets a solid background color

background-image

Sets an image as the background

background-repeat

Controls repeating of background image

background-position

Positions the background image

background-size

Scales the background image (cover/contain)



Typography & Text Styling

Making Text Readable & Attractive

Font Properties

CODE

```
p {  
  font-family: 'Arial', sans-serif;  
  font-size: 18px;  
  font-weight: bold;  
  font-style: italic;  
  line-height: 1.6;  
}
```

font-family

Sets the typeface used for text

font-size

Controls the size of the text

font-weight

Controls boldness (normal, bold, 100–900)

font-style

Italic or normal text style

line-height

Sets vertical spacing between lines

Text Styling Properties

CODE

```
h1 {  
  text-align: center;  
  text-decoration: underline;  
  text-transform: uppercase;  
  letter-spacing: 2px;  
  color: #2496ED;  
}
```

text-align

Aligns text: left, right, center, justify

text-decoration

Adds underline, overline, or strikethrough

text-transform

Changes case: uppercase, lowercase, capitalize

letter-spacing

Adjusts space between characters

color

Sets the text color

Borders & Spacing

Defining Edges & Whitespace

Border Properties

CODE

```
div {  
  border-width: 2px;  
  border-style: solid;  
  border-color: #2496ED;  
  border-radius: 10px;  
  
  /* shorthand */  
  border: 2px solid #2496ED;  
}
```

border-width

Sets thickness of the border

border-style

solid, dashed, dotted, double, none

border-color

Sets the color of the border

border-radius

Rounds the corners of an element

border (shorthand)

Combines width, style & color in one line

Margin & Padding

Both control spacing, but in different places relative to the border.

margin

Space OUTSIDE the border, between the element and its neighbors.

```
p {  
  margin: 10px;  
  margin-top: 5px;  
  margin-right: 10px;  
  margin-bottom: 5px;  
  margin-left: 10px;  
}
```

padding

Space INSIDE the border, between the border and the content.

```
p {  
  padding: 10px;  
  padding-top: 5px;  
  padding-right: 10px;  
  padding-bottom: 5px;  
  padding-left: 10px;  
}
```

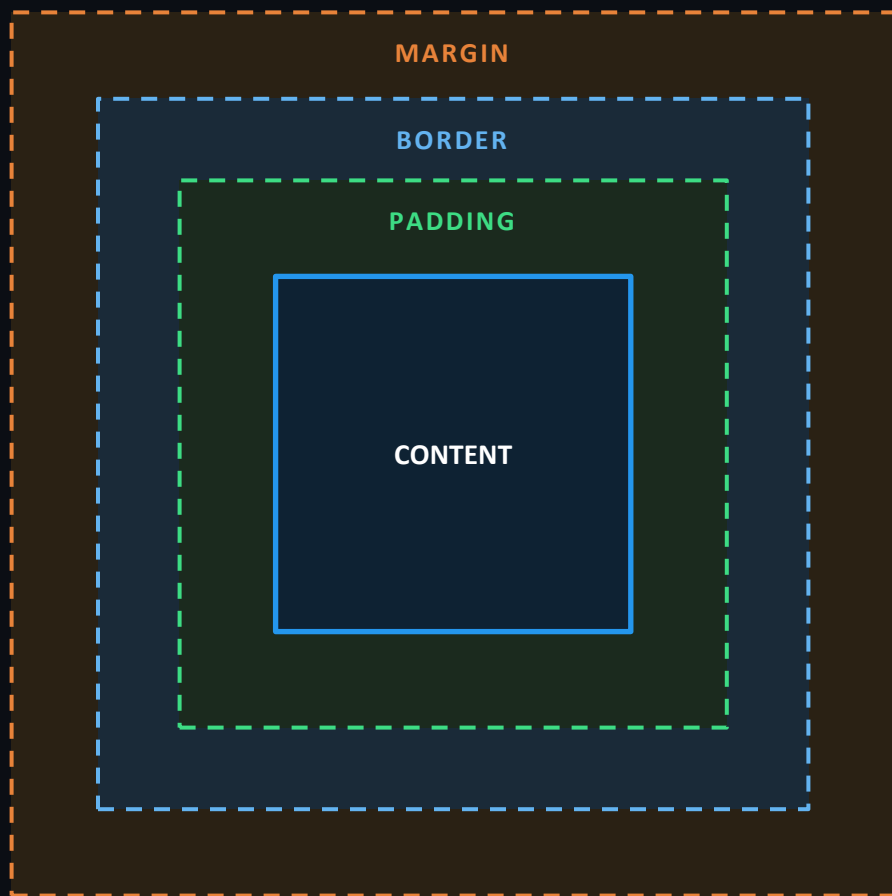


CSS Box Model

Understanding How Elements Take Up Space

The CSS Box Model

Every HTML element is a box made up of four layers.



Content

The actual text, image, or element content

Padding

Space between content and the border

Border

The line that wraps around padding & content

Margin

Space outside the border, separating elements

Box Model in Code

CODE

```
div {  
  width: 200px;  
  height: 100px;  
  padding: 20px;  
  border: 5px solid #333;  
  margin: 15px;  
  box-sizing: border-box;  
}
```

width / height

Sets the content box dimensions

box-sizing: border-box

Includes padding & border within the set width/height

Total element size

= content + padding + border + margin

Mini Project: Styled Profile Card

CODE

```
.profile-card {  
  width: 280px;  
  background-color: #ffffff;  
  border: 1px solid #ddd;  
  border-radius: 12px;  
  padding: 20px;  
  margin: 20px auto;  
  text-align: center;  
}  
  
.profile-card h2 {  
  font-family: 'Arial', sans-serif;  
  color: #2496ED;  
  text-transform: uppercase;  
}  
  
.profile-card p {  
  color: #555;  
  line-height: 1.5;  
}
```

CSS Concepts Used:

Selectors (class)

Background & border

Border-radius

Padding & margin

Typography (font, transform)

Box model sizing

Best Practices in CSS



Use external CSS files

Keeps HTML clean and styles reusable



Use classes over inline styles

Easier to maintain and reuse



Follow consistent naming

e.g. `.card-title` not `.ct1`



Group related properties

Keep font, color, spacing rules organized



Use shorthand properties

`border: 1px solid #000;` instead of 3 lines



Test across browsers

Check rendering & responsiveness

Today We Learned



Introduction to CSS

What CSS is & why it's used



Syntax & Selectors

Rules, properties, targeting elements



Colors & Backgrounds

Color formats & background properties



Typography

Fonts, text styling & spacing



Borders & Spacing

Border, margin & padding



Box Model

Content, padding, border, margin

Key Takeaway: CSS transforms plain HTML into structured, attractive, and user-friendly webpages.



Thank You!

Questions? I'm happy to help!

Practice Task:

“Style a Student Profile Card using CSS. Apply colors and a background, set custom typography, add borders with rounded corners, and use the box model (padding & margin) to space everything neatly.”

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